Chengdi 485 Communication Pressure Gague

MODBUS Protocol And Wiring Instructions

Brown: 12~24VDC, **Blue**:0V, White : 485A D+ , **Orange**: 485B D-, **Black**: Digital switching output

Data Format	Address	Funtion		Data		CRC
	8bit	8bit		n*2*8 bit		16bit
Single Frame Format	The default baud rate is 19200, with 1 start bit, 8 data bits, 1 stop bits, and no parity bit. The address of Pressure Gague can be adjusted from 1 to 247, with a default value of 01 (hexadecimal 01H).					
Function Code	03H: Read Multiple Registers 06H: Write Single Register			04H: Read the air pressure value 10H: Write Multiple Registers		

Register Address Description:

Address	Register pointing	Value		Corresponding Pressure Switch Ontology settings	Read /write
		High pressure	Low pressure		
0001H	Pressure value	-101~1000 corresponding to -0.101~1.000MPa	-1013~1000 corresponding to -101.3kPa~100kPa	NC	R
0002H (Follow the display screen)	Pressure value	-101~1000 corresponding to -0.101~1.000MPa	-1013~1000 corresponding to -101.3kPa~100kPa	NC	R
0003H	Switching output status	0~1 Corresponding to OFF, ON	0~1 Corresponding to OFF, ON	Switching output status	R
0010H	OUT target value	-100~1000 corresponding to -0. 100~1.000 MPa	-1010~1010kPa corresponding to -101.0~101.0kPa	Run mode: P-1	R/W
0011H	OUT1 target value upper limit value	out1_down_value~1000 corresponding to out1_down_value~1.000MP	out1_down_value~1010 corresponding to out1_down_value~101.0kPa	Run mode: Hi-1	R /W
0012H	OUT1 target value lower limit value	-100~out1_up_value corresponding to -0.100MPa~out1_up_value	-1010~out1_up_value corresponding to -101.0kPa~out1_up_value	Run mode: LO-1	R/W
0013H	OUT1 output mode	0~2 Corresponding to EASY,HYS,WCMP	0~2 Corresponding to EASY,HYS,WCMP	MENU mode: OUT1	R/W
0014H	Display color settings	0~3 Corresponding to R_ON,G_ON,RED,GREEN	0~3 Corresponding to R_ON,G_ON,RED,GREEN	MENU mode: COLO	R/W
0015H	Unit Setting Value	0~4 Corresponding to MPA ,KPA ,KGF ,BAR ,PSI	1~6 Corresponding to MPA ,KPA KGF ,BAR , PSI,MMHG,INHG	MENU mode: UNIT	R/W
0016H	Normally open and normally closed setting value	0~1 Corresponding to NO,NC	0~1 Corresponding to NO,NC	MENU mode: no.nc	R/W
0017H	Display value fine-tuning settings	-50~50 correspond to: displayed value=actual value * (1+adj1/100)		PRO mode: adj1	R/W
0018H	Micro value shielding function	0~30 correspond to: when the display value is between - adj2<display value<adj2, the display value is 0		PRO mode: adj2	R/W
0020H	Reset Settings	Writing 1 is the display reset operation When the barometer is connected to the atmosphere		Long press the up and down keys for 3 seconds	W

Instructions for changing the address of Pressure Gague:

1. Press the MODE key for 2 seconds to enter the MENU setting menu. Press the MODE key until the Addr menu appears, and adjust the address using the up and down keys.
2. Press the MODE key for 2 seconds to enter the MENU setting menu. Press the MODE key until the BAUD menu appears, and adjust the baud rate through the up and down keys, with 9600、19200、38400、115200 adjustable.
3. Press the MODE key for 2 seconds to enter the MENU setting menu. Press the MODE key until the Sbit menu appears, and adjust the Stop bit through the up and down keys, with1、2 adjustable.
4. Press the MODE key for 2 seconds to enter the MENU setting menu. Press the MODE key until the EocH menu appears, and adjust the Verification code through the up and down keys, with NONE、ODD or E_EN adjustable.

03H Command Example

Host send		
Drive Address		01H
Function Code		03H
Start Address	High-order	00H
	Low-order	10H
Quantity	High-order	00H
	Low-order	02H
CRC	Low-order	C5H
	High-order	CEH

Pressure switch return		
Drive Address		01H
Function Code		03H
Data Quantity		04H
Data 1	High-order	00H
	Low-order	00H
Data 2	High-order	00H
	Low-order	00H
CRC	Low-order	FAH
	High-order	33H

Host send:01 03 00 10 00 02 C5 CE

04H Command Example

Host send		
Drive Address		01H
Function Code		04H
Start Address	High-order	00H
	Low-order	01H
Quantity	High-order	00H
	Low-order	01H
CRC	Low-order	60H
	High-order	0AH

Pressure switch return		
Drive Address		01H
Function Code		04H
Data Quantity		02H
Data 1	High-order	00H
	Low-order	00H
CRC	Low-order	B9H
	High-order	30H

Host send:01 04 00 01 00 01 60 0A

06H Command Example

Host send		
Drive Address		01H
Function Code		06H
Start Address	High-order	00H
	Low-order	10H
Data Value	High-order	01H
	Low-order	90H
CRC	Low-order	89H
	High-order	F3H

Pressure switch return		
Drive Address		01H
Function Code		06H
Data Address	High-order	00H
	Low-order	10H
Data Value	High-order	01H
	Low-order	90H
CRC	Low-order	89H
	High-order	F3H

Host send:01 06 00 10 01 90 89 F3

10H Command Example

Host send		
Drive Address		01H
Function Code		10H
Start Address	High-order	00H
	Low-order	11H
Quantity	High-order	00H
	Low-order	02H
Data Quantity		04H
Data 1	High-order	00H
	Low-order	64H
Data 2	High-order	00H
	Low-order	00H
CRC	Low-order	72H
	High-order	B0H

Pressure switch return		
Drive Address		01H
Function Code		10H
Start Address	High-order	00H
	Low-order	11H
Quantity	High-order	00H
	Low-order	02H
CRC	Low-order	11H
	High-order	CDH

Host send: 01 10 00 11 00 02 04 00 64 00 00 72 B0

Pressure Sensor Catalog

MS30 series

NPN/PNP

MS31 series

Analog output

MS32 series

RS485
communication

MS33 series

Corrosion preventive

MS36 series

IP65 waterproof

MS38 series

High-precision
RS485 communication

MS39 series

Micro pressure
RS485 communication

MS39 series

Micro pressure
analog output

MS20 series

Ultra thin series

MKV series

Split type

MT1 series

NEW
IP65 Waterproof

MT20 series

Fluid series

Chengdi

Pressure Sensor Features

Advantages 1

13 years of technical accumulation,3000,000+ sets cumulative sales,supporting OEM/0DM customization.

High Precision 2

Accurate 0.1kpa/0.01kpa/0.001kpa optional， Support unit switching: MPa, kPa, KGF, bar,PSI, mmHg,inHg (slightly different for different models).

Output Mode 3

NPN/PNP Switch output,Analog output 1-5V/4-20mA,RS485 communication,Multiple output modes to choose.

High Protection 4

IP65 Waterproof,Sus304/Sus316L Corrosion resistant, suitable for applications with high protection level requirements.

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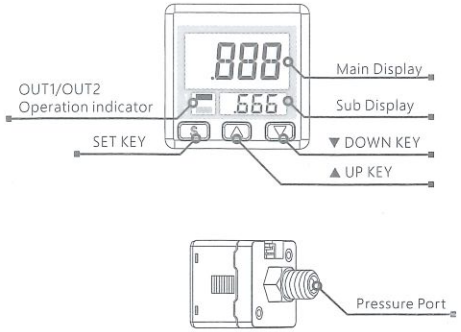
INSTRUCTION MANUAL

Digital Pressure Switch MS Series

◆◆◆WARNING◆◆◆

① Do not use corrosive or flammable gas or liquid with this product.
② Please use within the rating pressure range. Do not apply pressure beyond recommended maximum withstand pressure, permanent damage to the pressure sensor may occur.
③ Do not drop, hit or allow excessive shock. Even if switch body appears undamaged, internal components may be broken and can cause malfunction.
④ Turn power off before connecting wiring. Wrong wiring or short circuit will damage and / or cause malfunction.
⑤ Do not use in environment containing steam or oil vapor.
⑥ This product is not explosion-proof rated. Do not use in atmosphere containing flammable or explosive gases.
⑦ Wiring for pressure sensor should avoid power source line and high voltage line. If use in the same circuit, noise may cause malfunction.

A. PANEL DESCRIPTION



B. ORDERING INFORMATION

MS 30 C 001 A

Product type
30-Economic series
31-Stangdard series
32-Communication series
38-High precision series
39-Micro pressure series

Pressure type
C:-101.3~101.3kPa
P:-0.101~1.000MPa
CA:-500~500Pa
CB:-1000~1000Pa
CE:-10.00~10.00kPa

Output type
001 -1NPN Output
002 -1PNP Output
010 -2NPN Output+1analog(1-5V)
011 -2NPN Output+1analog(4-20mA)
020 -2PNP Output+1analog(1-5V)
021 -2PNP Output+1analog(4-20mA)
060 -1NPN Output+1RS485
061 -1PNP Output+1RS485

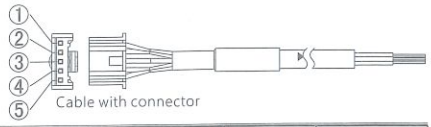
Interface Type
A - R1/8",M5
M - M5

Optinal Parts for purchasing separately

1. Metal mounting bracket - L style: XP-BT-01
2. Metal mounting bracket - Z style: XP-BT-02
3. Panel adapter(have protective lid): XP-PA-01
4. Cable - 3 wires: XP-CA-01
5. Cable - 5 wires: XP-CA-02

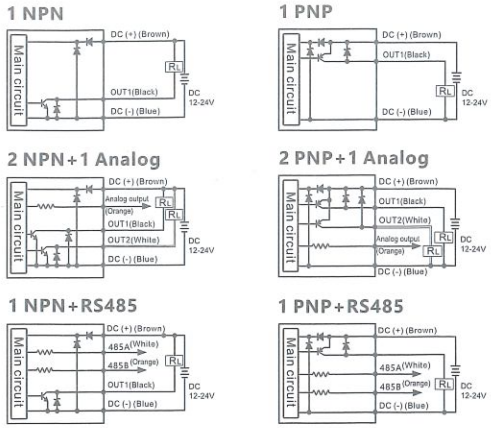
C. Conection connector pin arrangement

Note: Do not pull by holding the cable without pressing the release lever, as this can cause cable break or connector break.



Pin NO.	Terminal name (MS 31 Series)	Terminal name (MS 30 Series)	Terminal name (MS 32/38/39 Series)
① Brown	DC+	DC+	DC+
② Black	OUT1	OUT1	OUT1
③ White	OUT2		485A
④ Orange	AIO		485B
⑤ Blue	DC-	DC-	DC-

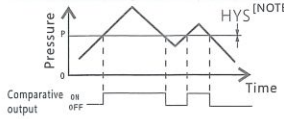
D. OUTPUT CIRCUIT WIRING DIAGRAMS



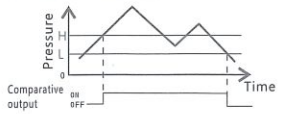
E. OUTPUT TYPE

The EASY mode, hysteresis mode or window comparator mode can be selected as the output mode for comparative output 1 and comparative output 2. Please refer to <I.NORMAL mode> for details.

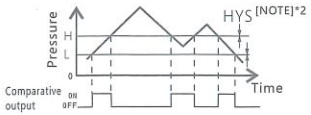
EASY mode



Hysteresis mode



Window comparator mode

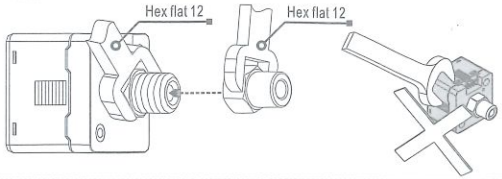


【NOTE】
*1. In the EASY mode, P-1 / P-2 is displayed on the sub-display, Hi-1 / Hi-2 / Lo-1 / Lo-2 is displayed in the other mode.
*2.Please refer to <Hysteresis fixed value selection> in <PRO setting> for setting.

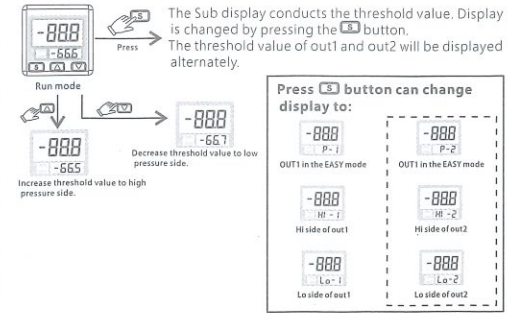
In this mode, the ON or OFF state of the comparative output is controlled with a pressure in the set range.

F. PIPING

When connecting a commercial coupler to the pressure port, attach a 12mm spanner to the pressure port's hexagon section to fix the port, and then tighten with a tightening torgue of 9.8 N·m or less(M5 female: 1N·m or less). The commercial coupler or pressure port section will be damaged if the tightening torque is excessive. Wrap sealing tape around the coupler when connecting to prevent leaks.

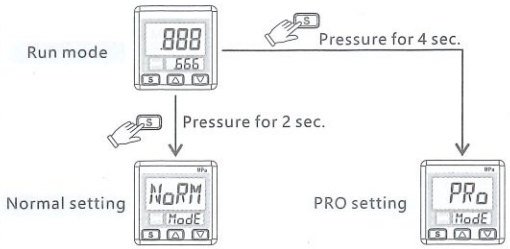


G. RUN MODE

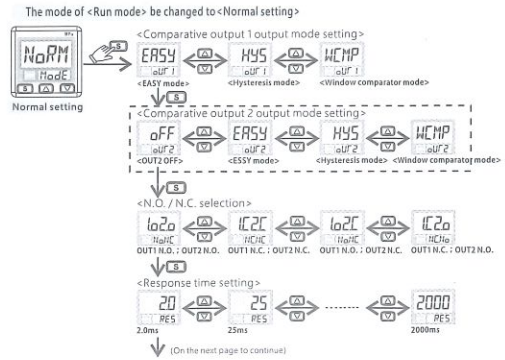


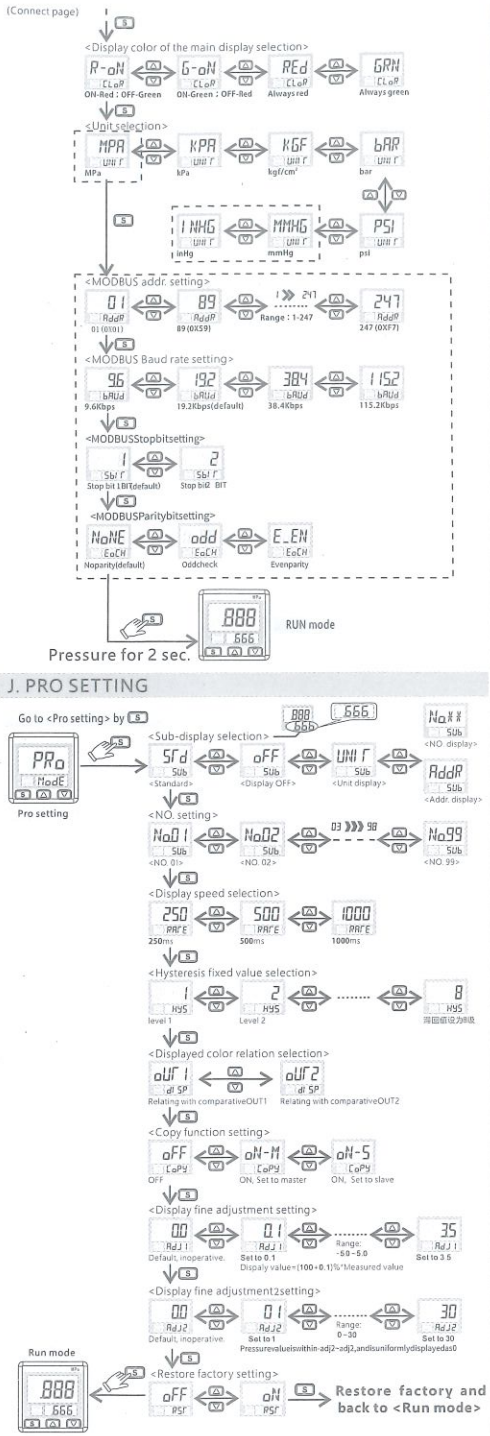
The Sub display conducts the threshold value. Display is changed by pressing the S button.
The threshold value of out1 and out2 will be displayed alternately.

H. MENU SETTING MODE



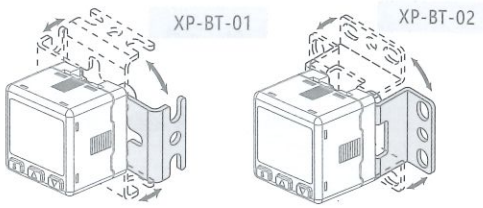
I. NORMAL SETTING



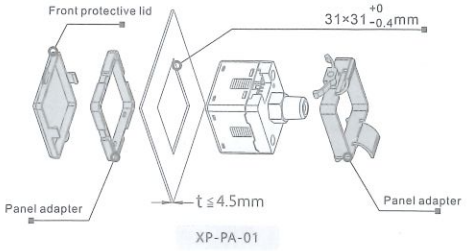


K. MOUNTING

1 - Mounting bracket



2 - panel mounting



L. SUPERSPEED MODE

By the SUPERSPEED mode, response time can up to 1ms !

【NOTE】
JUST for MS31 series

Press [S] + [△] button at the same time on the <Run mode> until display show as:

SUPERSPEED ON

SUPERSPEED OFF

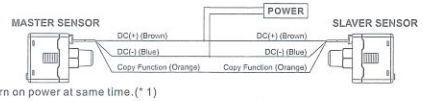
Release the buttons to end setting.

M. COPY FUNCTION SETTING

◎ Copy function setting can use the master sensor to copy the pressure value to the slave sensors.
◎ Before copying, please confirm the model of pressure sensor. The function cannot use in difference mode.
◎ The copy function only can be one-to-one.

【SETTING STEP】

1. Please set the copy function to [on] or [onL] to be on copy condition by master sensor. Please refer the copy setting of (H) advance setting mode.
2. Turn power off to both sensor.
3. Refer the connection way with the master and slaver sensor as followings.



4. Turn on power at same time.(* 1)
5. Wait 5 sec., when finishing to convey the data,
   the master sensor display (mutually display) [CPY] ⟷ [Gad] (display reciprocal)
   the slaver sensor displ [Slv] ⟷ [Gad] ay (mutually display) (display reciprocal)
6. When convey the data is fail,
   (Master) sensor displays [CPY] ⟷ [Gad] (display reciprocal) (Slaver) sensor displays [Err] (* 2)
7. Turn off power and remove the wire connection. If no remove the wire connection, the sensor would be broken.

★If require to copy another slave sensor, please repeat the step ③ to ⑤.
★Only for KP43□-02-□ and KP43□-04-□ have this setting selection.

【NOTE】

*1. If turn on power is not synchronization, the data cannot be copied.
*2. When the data conveys fail, please confirm the wire connection. Then repeat the step ③ to ⑤.

How to cancel the copy mode by master sensor:

◎ When the master sensor display [CPY] ⟷ [Gad] (display reciprocal), Please [▼] button to leave the copy mode.

N. SPECIFICATIONS

Model		MS30/31/32P (High Pressure)	MS30/31/32C (Low Pressur)e	MS38P (High Precision And High Pressur)e	MS38C (High Precision And Low Pressure)	MS39 (Slight Pressure)
Rated Pressure Range		-0.101~1.000MPa	-101.3~101.3kPa	-0.101~1.000MPa	-101.3~101.3kPa	-10~10kPa
Display Pressure Range		-0.101~1.000MPa	-101.3~101.3kPa	-0.101~1.000MPa	-101.3~101.3kPa	-10~10kPa
Withstang Pressure		1.5MPa	500kPa	1.5MPa	500kPa	20kPa
Applicable Fluid		Filtered Air Non-corrosive/Non-flammable Gas				
Power Supply		12~24V DC±10%				
Current Consumption		≤45mA (Withnoload)				
Set Pressure Resolution	kPa	1	0.1	0.1	0.01	0.001
	Mpa	0.001	-	-	-	-
	kgf	0.01	0.001	-	-	-
	bar	0.01	0.001	-	-	-
	psi	0.1	0.01	-	-	-
	mmHg	-	0.1	-	-	-
	inHg	-	1	-	-	-
Switch Output		NPN:openCollector Maxload : 100mA MaxsupplyVoltage : 30V DC ResidualVoltage : ≤1.5V		PNP:openCollector Maxload : 100mA MaxsupplyVoltage : 24V DC ResidualVoltage : ≤1.5V		
Analog Output		VoltageOutput : Output Voltage:1~5v ±2.5%F.S. Linearity ±1%F.S. Output Impendance:about 1KΩ		CurrentOutput : Output Voltage:1~5v ±2.5%F.S Linearity ±1%F.S. Output Impendance:Min 5Ω 0 & Max:600Ω		
		Note: Only MS31 series have analog output function.				
Indicator Accuracy		±1%F.S. ±1digits(Ambienttemperature25 ±10 ℃)				
Repeatability		±0.2%F.S ±2digits				
Response Time		2.0ms, 50ms,100ms, 250ms,500ms, 200 0ms Selectable Note: The fastest 1ms for MS31 analog output				
LCD Display		3 Color (Red/Green/Orange) Display (Sampling Rate:4 Times/1sec.				
Temperature Characteristic		1%F.S. of detected pressure (25℃) at temperature range of 0~50℃				
Working Environment		0~50℃ No condensatio or freezing);35~85%RH(No condensation)				
Storage Environment		-10~50℃ (No condensatio or freezing);35~85%RH(No condensation)				
Protection Level		IP40(IEC)				
Pressure Port		A,R1/8+M5 B,G1/8+M5 C,NPT1/8+M5 M,M5				
Weight		Approximate 80g(With lead wrie)				
Accessories		A cable with connertor & an instruction manual				

【Note】

1. Output type and product model-one correspondence, please refer to "Order Guide" for details
2. MS32/MS38/MS39 series have RS485 interface, support MODBUS RTU mode, Please refer to "Product Manual" for its operation
3. Turn on SUPERSPEED mode to support 1ms response speed, about this mode Please refer to "Product Manual" for opening and closing
4. For the setting of response speed, please refer to "Product Manual"

O. MODBUS

Data format	Address	Funtion	Data	CRC
	8bit	8bit	n*8bit	16bit
Single	1 start bit, 8 data bits, 1stop bits, no parity.			

REGISTER ADDRESS

addr.	Point to	Value High	Value Low	Corresponding to the set	Limits
0001H	Threshold value	-100~1000 ↕ -0.101~1.000MPa	-1013~1013 ↕ -101.3kPa~101.3kPa	NC	R
0010H	OUT1	-100~1000 ↕ -0.100~1.000MPa	-1010~1010 ↕ -101.0~101.0 kPa	-888 P-1	R/W
0011H	OUT1 Hi-side	Lower limit 1000 ↕ Lowerlimit pressurevalue ~1.000MPa	Lowerlimit~1010 ↕ Lowerlimit pressurevalue ~101.0kPa	-888 Hi-1	R/W
0012H	OUT1 Lo-side	-100~Upperlimit ↕ -0.100MPa~Upperlimit pressurevalue	-1010~Upperlimit ↕ -101.0kPa~Upperlimit pressurevalue	-888 Lo-1	R/W
0013H	OUT1 mode	0~2 ↕ EASY,HYS, WCMP	0~2 ↕ EASY,HYS, WCMP	EASY oUT1 HYS oUT1 WCMP oUT1	R/W
0014H	Display color	0~3 ↕ R_ON,G_ON, RED,GREEN	0~3 ↕ R_ON,G_ON, RED,GREEN	R-oN CLoR G-oN CLoR REd CLoR GRN CLoR	R/W
0015H	Unit	0~4 ↕ MPa,kPa, kgf/cm²,bar,psi	1~6 ↕ kPa,kgf/cm², bar,psi,mmHg, inHg	MPA unit kPA unit kGF unit bAR unit PSI unit MMHG unit I NHG unit	R/W
0016H	N.O. N.C.	0~1 ↕ NO. NC.	0~1 ↕ NO. NC.	INo NoNC INC NoNC	R/W

(On the next page to continue)

(Connect page)

CODE	FUNCTION
03H	Read multipe registers
04H	Read the current pressure
06H	Write a single register
10H	Write multipe registers

Abnormal code	Descriptions
01H	code error
02H	Illegal data address, out of address range or read-write permission
03H	Illegal data, the number of data exceeds the limit or error count
04H	Parameter setting error, parameter setting value writing out of setting range

P. ZERO POINT SETTING

Press [△] + [▽] at the same time until show as follow:

Release the button to end zero setting.

Q. KEY LOCK/UNLOCK

Press [S] + [▽] at the same time until show as follow:

LOCK

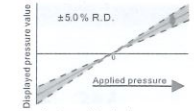
UNLOCK

Release the button to end setting

R. DISPLAY FINE ADJUSTMENT FUNCTION

<Pro setting>

This function eliminates slight differences in the output values and allows uniformity in the numbers displayed.
Displayed values of the pressure sensor can be calibrated to within ±5.0% R.D.



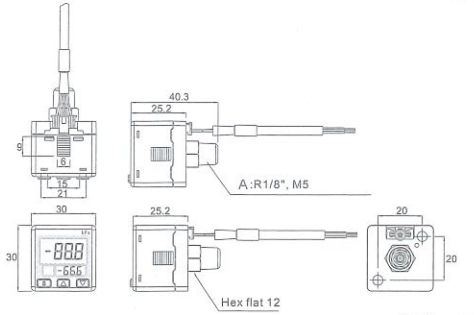
R.D. (Real Detect)
【NOTE:】 1. setting resolution is ±0.1% R.D.
2. The signal would be changed with analog output after adjusting.

— :Factory setting display value set prior to shipment
▭ :Display calibration range

S. EERROR CODE

Error Type		Errcode	Error Condition	Troubleshooting
Excess load current error	out1	Er1	Output 1 load current is more than 125 mA	Turn power off and check the cause of overload current or lower the current load under 125 mA, then restart.
	out2	Er2	Output 2 load current is more than 125 mA	
Residual pressure error		Er3	During zero reset, ambient pressure is over ±3% F.S.	Change input pressure to ambient pressure and perform zero reset again.

T. DIMENSIONS



Unit: mm